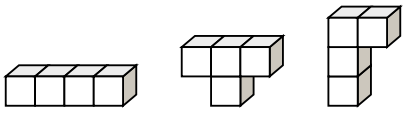
Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q23 (v1)

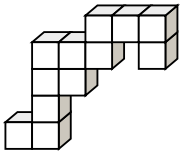
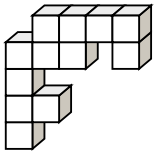
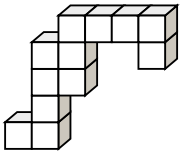
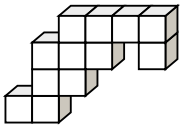
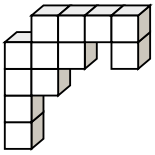
Fit given pieces into a shape: start where a piece fits only one way, then look at what is left

Part A: Each question is worth 3 points

1. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over.

Which of these figures **can** he make?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

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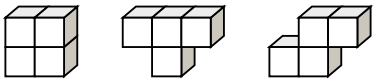
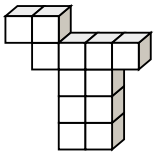
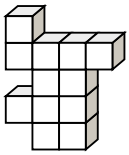
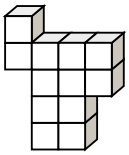
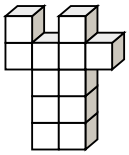
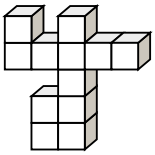
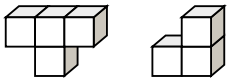
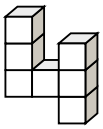
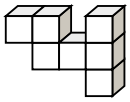
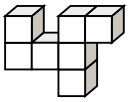
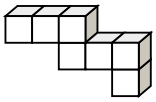
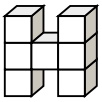


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

3. Kanga wants to build a flat figure out of these 2 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over.

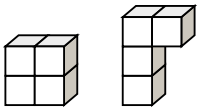
Which of these figures **can** he make?

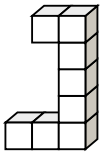
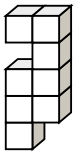
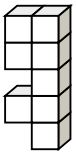
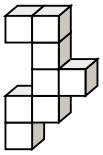
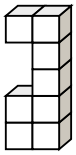


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

4. Kanga wants to build a flat figure out of these 2 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over.

Which of these figures **can** he make?

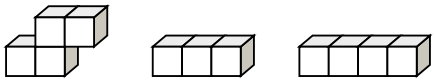


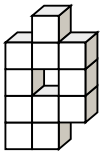
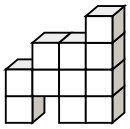
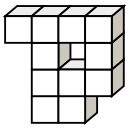
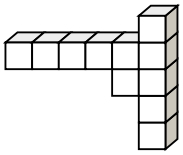
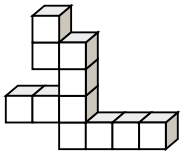
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part B: Each question is worth 4 points

5. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over.

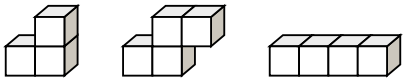
Which of these figures can he **not** make?

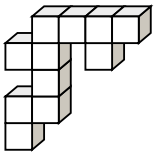
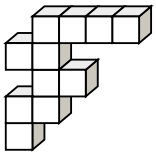
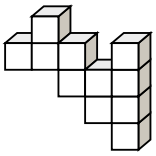
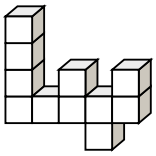
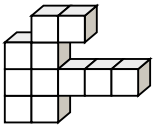


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over.

Which of these figures can he **not** make?

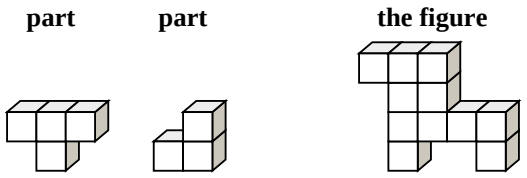


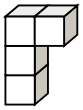
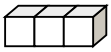
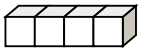
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part C: Each question is worth 5 points

7. Kanga built this flat figure out of 3 parts made of cubes, turned or flipped. Two of the parts are shown next to it.

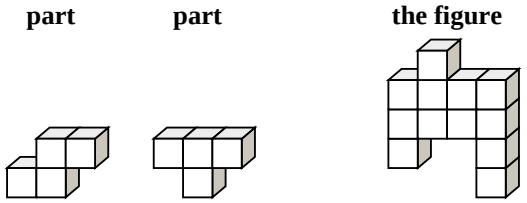
Which part is the third one?

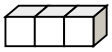
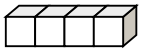


A.  ***B.*** ***C.***  ***D.***  ***E.***

8. Kanga built this flat figure out of 3 parts made of cubes, turned or flipped. Two of the parts are shown next to it.

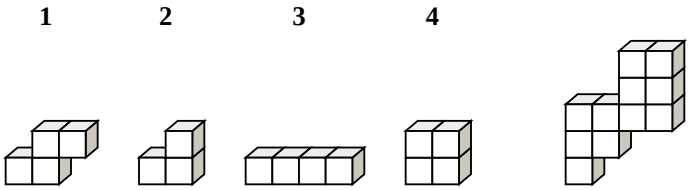
Which part is the third one?



A. ***B.*** ***C.*** ***D.***  ***E.*** 

9. Kanga has these 4 parts made of cubes (numbered 1 to 4). He builds the flat figure on the right with **three** of them, turned or flipped.

Which part is left over?

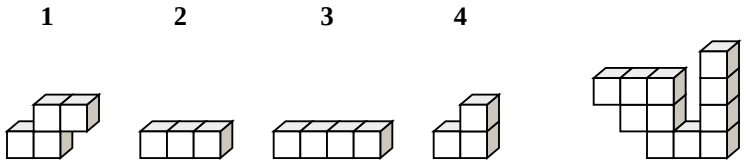


A. part 1 ***B.*** part 2 ***C.*** part 3

D. part 4 ***E.*** none, he needs all 4

10. Kanga has these 4 parts made of cubes (numbered 1 to 4). He builds the flat figure on the right with **three** of them, turned or flipped.

Which part is left over?



A. part 1 ***B.*** part 2 ***C.*** part 3

D. part 4 ***E.*** none, he needs all 4

Answer Key

1. **E**	2. **D**	3. **B**	4. **E**	5. **A**	6. **A**
7. **A**	8. **E**	9. **A**	10. **D**		

Solutions

1. E — Every figure has 12 cubes, so count is not enough. Start with a cube that sticks out: only one part can fill it. Only one figure can be filled completely. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · can_make]

2. D — Every figure has 12 cubes, so count is not enough. Start with a cube that sticks out: only one part can fill it. Only one figure can be filled completely. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · can_make]

3. B — Look for one part inside the figure; what is left must be the other part. That works only for figure B. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · two_parts]

4. E — Look for one part inside the figure; what is left must be the other part. That works only for figure E. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · two_parts]

5. A — Try to fill each figure with the parts, starting at a corner or a cube that sticks out. Four figures can be filled; in figure A there is always a gap that no part fits. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · cannot_make]

6. A — Try to fill each figure with the parts, starting at a corner or a cube that sticks out. Four figures can be filled; in figure A there is always a gap that no part fits. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · cannot_make]

7. A — The figure has 11 cubes and the two parts have 7, so the third part has 4 cubes. Put the two parts into the figure; what is left has the shape of part A. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · missing_part]

8. E — The figure has 12 cubes and the two parts have 8, so the third part has 4 cubes. Put the two parts into the figure; what is left has the shape of part E. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · missing_part]

9. A (part 1) — The figure has 11 cubes and the parts have 15 together, so the left-over part has 4 cubes. Then check which three parts really fill the figure: part 1 is left over. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · left_over]

10. D (part 4) — The figure has 11 cubes and the parts have 14 together, so the left-over part has 3 cubes. Then check which three parts really fill the figure: part 4 is left over. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · left_over]